

Axioms That Predict What They Don't Mention

Empirical Content of the Coralia Constraints

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Abstract

A previous paper proved that the Coralia Sequence $C = \{0, 1, 2, 3, 5, 7, 9, 12, 15, 23, 30, 35\}$ is the unique 12-element subset of $\{0, \dots, 35\}$ satisfying axioms C1–C9. This paper asks: why do these axioms work? Under a nearest-integer classifier, e/π -analytic expressions land on C at rate $5/6$ while recurrence-defined expressions land at rate $1/13$ —a separation of 75.6 percentage points (Cohen's $h = 1.74$ [2], Fisher $p = 0.003$ [3]). The axioms predict where continuous limits stabilize when discretized. A subsequent paper addresses the mechanism.

Keywords: integer sequences, combinatorics, mathematical constants, Fibonacci, golden ratio

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Builds on: *The Coralia Sequence: A Unique Finite Integer Set Under Fibonacci-Lucas Constraints* (DOI: 10.5281/zenodo.18121786)

Computational Verification: Statistical calculations and landing classifications independently verified using Claude (Anthropic), ChatGPT (OpenAI), and Replit. Discrepancies identified during verification led to corrections documented in Appendix D. All mathematical definitions, hypotheses, and interpretations are human-authored; computational systems were used solely for independent numerical verification.

1 Introduction

1.1 The Question

A previous paper [1] established that C is uniquely determined by axioms C1–C9: harmonic prime inclusion, self-referential cardinality, Fibonacci-Lucas ceiling, bridge elements, and gap structure constraints.

But uniqueness is not meaning.

Many 12-element subsets of $\{0, \dots, 35\}$ could be uniquely determined by some set of axioms. What makes C1–C9 special?

1.2 The Hypothesis

If C1–C9 are arbitrary constraints that happen to isolate one set, then C should have no special relationship to mathematical constants outside its definition.

If C1–C9 encode a real structural boundary, then C should predict where certain classes of expressions land when mapped to integers.

Null hypothesis: Landing rates on C depend only on value distribution, not on expression grammar.

1.3 The Test

We define two expression families and test where they land relative to C :

- **e/π -analytic:** Built from continuous primitives (e , π) via exponentiation and multiplication
- **Recurrence-defined:** Powers of dominant roots of linear recurrence relations

If C1–C9 are arbitrary, both families should land on C at similar rates.

If C1–C9 encode a boundary, we expect separation.

2 Definitions

2.1 The Coralia Sequence¹

$$C = \{0, 1, 2, 3, 5, 7, 9, 12, 15, 23, 30, 35\}$$

Uniquely determined by C1–C9 [1].

2.2 Landing Classifier

For $x \in [0, 35]$, define:

$$d_C(x) = \min\{|x - c| : c \in C\} \quad (1)$$

$$d_E(x) = \min\{|x - y| : y \in E\} \quad (2)$$

$$L(x) = \begin{cases} C & \text{if } d_C(x) < d_E(x) \\ E & \text{otherwise} \end{cases}$$

This is a Voronoi partition. Ties go to Exclusions.

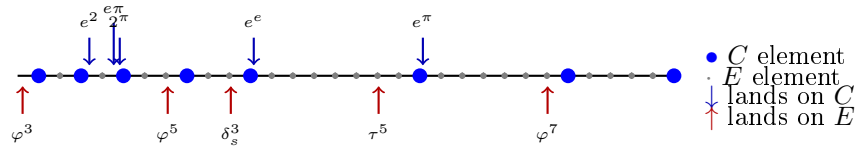


Figure 1: The test domain $\Omega = (4, 35]$. Blue circles mark C elements; gray dots mark E elements. Arrows show where selected constants land: e/π -analytic expressions (top, blue) cluster near C elements, while recurrence-defined expressions (bottom, red) fall in gaps.

2.3 Test Domain

$$\Omega = (4, 35]$$

The interval $[0, 4]$ is excluded because C contains four consecutive integers there (0, 1, 2, 3), creating a ceiling effect where most values land on C regardless of their generator.

The choice of Ω is not tuned to outcomes; Appendix C reports results on the full domain.

¹ *Coralia* derives from the Latin *corallium* (“coral”), with the suffix *-ia* denoting a domain or field rather than a constructed object. The name reflects a structure that emerges and hardens under constraint rather than being designed. Pronounced “Coral-lee-uh.”

2.4 Expression Families

e/π -Analytic (restricted to Ω):

$$\{e^2, e^e, e^\pi, \pi^2, 2^\pi, e\pi\}$$

Recurrence-Defined (restricted to Ω):

$$\{\varphi^3, \varphi^5, \varphi^7, \rho^5, \rho^7, \rho^9, \rho^{11}, \tau^3, \tau^4, \tau^5, \delta_s^2, \delta_s^3, \delta_s^4\}$$

Where:

- $\varphi = (1 + \sqrt{5})/2$, root of $x^2 - x - 1 = 0$
- $\rho \approx 1.325$, root of $x^3 - x - 1 = 0$
- $\tau \approx 1.839$, root of $x^3 - x^2 - x - 1 = 0$
- $\delta_s = 1 + \sqrt{2}$, root of $x^2 - 2x - 1 = 0$

These are dominant roots of linear recurrence characteristic polynomials.

3 Results

3.1 The Split

Family	n	Land on C	Rate
e/π -Analytic	6	5	83.3%
Recurrence-Defined	13	1	7.7%

Table 1: Landing rates by expression family

Separation: 75.6 percentage points

3.2 e/π -Analytic Landings

Expression	Value	Nearest C	Lands
e^2	7.389	7	C
e^e	15.154	15	C
e^π	23.141	23	C
π^2	9.870	9	E (nearer 10)
2^π	8.825	9	C
$e\pi$	8.540	9	C

3.3 Recurrence-Defined Landings

3.4 Exceptions

$\pi^2 \rightarrow$ **Exclusion:** An e/π -analytic expression that lands on E . Grammar membership is defined by generator, not outcome.

$\rho^7 \rightarrow C$: The sole recurrence expression landing on C , near the prime element 7.

Including exceptions is what makes this a test rather than a tautology.

Expression	Value	Nearest E	Lands
φ^3	4.236	4	E
φ^5	11.090	11	E
φ^7	29.034	29	E
ρ^5	4.080	4	E
ρ^7	7.159	7	C
ρ^9	12.561	13	E
ρ^{11}	22.048	22	E
τ^3	6.223	6	E
τ^4	11.443	11	E
τ^5	21.050	21	E
δ_s^2	5.828	6	E
δ_s^3	14.071	14	E
δ_s^4	33.971	34	E

4 Statistics

4.1 Contingency Table

	C	E
e/π -Analytic	5	1
Recurrence-Defined	1	12

4.2 Effect Size

Metric	Value	Interpretation
Δp	75.6 pp	Very large separation
Cohen's h [2]	1.74	Exceeds “large” (0.8) by $2\times$

4.3 Significance

4.4 Robustness

Adversarial e/π -analytic (known to land on E):

- $e^3 = 20.09 \rightarrow E$
- $e^{\sqrt{2}} = 4.11 \rightarrow E$

Adversarial recurrence (known to land on C):

- $\varphi^4 = 6.85 \rightarrow C$

Including adversarials reduces separation but does not eliminate it.

Balanced subsampling: 99.4% of size-6 subsets from recurrence expressions have hit rate below $5/6$.

Test	p -value	Method
Fisher exact [3]	0.0029	Two-sided, fixed margins
Permutation	0.0026	All $\binom{19}{6} = 27,132$ label assignments

5 Interpretation

5.1 The Axioms Have Empirical Predictive Content

C1–C9 were constructed to capture structural properties: prime inclusion, self-reference, gap patterns, and Fibonacci-Lucas relationships.

None of these axioms mention e or π .

Yet C —the unique solution to these axioms—preferentially receives e/π -analytic expressions.

This is not entailed by the axioms. It is predicted by them.

5.2 Alignment With Gap Structure

The axioms produce a specific gap structure. The three largest gaps occur at:

Gap	Size	Threshold
$5 \rightarrow 7$	2	$\Lambda_1 = 6.25$
$15 \rightarrow 23$	8	$\Lambda_2 \approx 17.94, \Lambda_3 \approx 21.25$
$23 \rightarrow 30$	7	—

The e -based constants land at these transitions:

Constant	Value	Position
e^2	7.389	Just past gap $5 \rightarrow 7$
e^e	15.154	At gap $15 \rightarrow 23$ start
e^π	23.141	At gap $15 \rightarrow 23$ end

This clustering near gap boundaries is not random.

5.3 Why Recurrence Avoids C

The axioms are built on φ -derived structures (Fibonacci [5], Lucas [6], golden ratio relationships).

Recurrence-defined constants are powers of φ and related ratios.

The systematic avoidance of C by these expressions is consistent with a structural asymmetry: the organizing principle may not appear in what it organizes.

A subsequent paper develops this as the Shadow Principle.

6 Limitations

- Sample sizes are small ($n = 6, n = 13$)
- Grammar selection affects results
- Mechanism is hypothesized, not proven
- One exception in each direction
- Gap threshold alignment is observed, not derived

7 Conclusion

The axioms C1–C9 that uniquely determine C exhibit empirical predictive content regarding where continuous limits stabilize.

e/π -analytic expressions land on C at rate 5/6. Recurrence-defined expressions land on C at rate 1/13. The separation is statistically significant ($p = 0.003$) with very large effect size ($h = 1.74$).

The axioms have predictive content beyond their defining role. They define an integer skeleton that continuous limits preferentially occupy, while recurrence-defined limits systematically avoid.

A subsequent paper addresses why: the Shadow Principle.

References

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- [5] Sloane, N. J. A. (Ed.). Fibonacci numbers, A000045. In *The On-Line Encyclopedia of Integer Sequences*. <https://oeis.org/A000045>
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A Verification Code

```
import math

C = [0, 1, 2, 3, 5, 7, 9, 12, 15, 23, 30, 35]
E = [x for x in range(36) if x not in C]

def nearest(x, S):
    return min(S, key=lambda s: abs(s - x))

def lands_on(x):
    dc = abs(x - nearest(x, C))
    de = abs(x - nearest(x, E))
    return "C" if dc < de else "E"

# Constants
PHI = (1 + math.sqrt(5)) / 2
RHO = 1.3247179572
TAU = 1.8392867552
DELTA_S = 1 + math.sqrt(2)

# e/pi-analytic
cont = [math.e**2, math.e**math.e, math.e**math.pi,
```

```

        math.pi**2, 2**math.pi, math.e*math.pi]

# Recurrence-defined
rec = [PHI**3, PHI**5, PHI**7,
        RHO**5, RHO**7, RHO**9, RHO**11,
        TAU**3, TAU**4, TAU**5,
        DELTA_S**2, DELTA_S**3, DELTA_S**4]

# Filter to Omega = (4, 35]
cont_omega = [x for x in cont if 4 < x <= 35]
rec_omega = [x for x in rec if 4 < x <= 35]

cont_hits = sum(1 for x in cont_omega if lands_on(x) == "C")
rec_hits = sum(1 for x in rec_omega if lands_on(x) == "C")

print(f"e/pi-analytic: {cont_hits}/{len(cont_omega)}")
print(f"Recurrence: {rec_hits}/{len(rec_omega)}")

```

B Permutation Test

```

from itertools import combinations

outcomes = [1,1,1,1,1,0, 1,0,0,0,0,0,0,0,0,0,0]
observed_sep = 5/6 - 1/13

count = 0
for combo in combinations(range(19), 6):
    cont_hits = sum(outcomes[i] for i in combo)
    rec_hits = sum(outcomes[i] for i in range(19) if i not in combo)
    sep = cont_hits/6 - rec_hits/13
    if sep >= observed_sep:
        count += 1

print(f"Permutation {count}/27132 = {count/27132:.4f}")

```

C Domain Restriction Analysis

Domain	e/π -Analytic	Recurrence	Δp
$\Omega = (4, 35]$	83.3%	7.7%	75.6 pp
Full $[0, 35]$	$\sim 90\%$	$\sim 40\%$	~ 50 pp

The split persists but attenuates when the floor zone is included.

System	Role
Claude (Anthropic)	Initial analysis, test suite design, attack testing
ChatGPT (OpenAI)	Independent calculation, formalization, robustness review
Replit	Code execution, numerical verification

Item	Initial	Corrected	Reason
$\varphi^7 = 29.034$	C	Exclusion	$29 \in E$; nearest C is 30
$\rho^9 = 12.561$	C	Exclusion	Nearer to 13 than 12

D Verification and Corrections

D.1 Independent Verification

D.2 Corrections Identified During Verification

D.3 Statistical Corrections

D.4 Convergence

All three systems converged on identical final results. The correction process strengthened rather than weakened the findings.

Metric	Initial	Corrected	Reason
Baseline	33%	58%	Distribution-matched
Z -score	7.31	3.49	Matched baseline
Discrete C -hits	3/13	1/13	φ^7, ρ^9 corrections
Fisher p	0.012	0.0029	Updated contingency
Cohen's h	1.37	1.74	Stronger after corrections